Power System Controls Engineer – Data Centers

Jacobs · Dallas, TX

Job ID (this collection)	CONSULT-054
Employer	Jacobs
Location(s)	Dallas, TX
Career stage	Experienced (5+ years asked)
Location group	DFW area
Majors	Electrical Engineering, Computer Engineering
Consulting area	Engineering design & infrastructure consulting
Pay listed	$145,200 – $199,700 per year
Recruiting status when checked	Open when checked: the employer page showed the posting on Oct 28, 2026.
Employer page	https://careers.jacobs.com/en_US/careers/JobDetail/x/46805
Captured	October 28, 2026

> Saved on October 28, 2026 from Jacobs's own careers page. Everything below the line is the employer's own wording, copied word for word and checked against the live page; only the layout was changed for printing. Live listings change and close, so check the employer page before applying.

Full job posting

Location Boston, Massachusetts, United States

Capabilities Electrical

Office Setup Hybrid

Job ID #46805

Industry Enterprise Functions

At Jacobs, we're challenging today to reinvent tomorrow by solving the world's most critical problems for thriving cities, resilient environments, mission-critical outcomes, operational advancement, scientific discovery and cutting-edge manufacturing, turning abstract ideas into realities that transform the world for good.

Your impact

At Jacobs, we bring engineering expertise, fresh thinking, and collaboration to complex energy challenges. Join us in developing power systems that support the next generation of data centers and advanced facilities while advancing reliability and resilience.

Your impact

Jacobs' Energy & Power business is seeking a Power System Controls Engineer to support conceptual planning, design, modeling, integration, and validation of Data Center controls ecosystem including battery energy storage systems (BESS), UPS, and other controllable equipment with a particular focus on AI and hyperscale facilities.

You'll help clients setup technical and functional specifications, define how BESS operates alongside utility supplies and onsite generation to comply with large load interconnection, performance and other applicable technical requirements.

Working with Jacobs' regional and global teams, data-center owners, equipment manufacturers, utilities, and EPC contractors, you'll support projects from feasibility and conceptual design through procurement, detailed engineering, commissioning, and operational performance assessment.

Your responsibilities will include:

- Define and establish overall AI data center plant control hierarchy and support establishing functional/performance specifications, acceptance criteria for load swings, faults, trips etc.;
- Work closely with state-of-the-art controls and inverter manufacturers to stay on top of all latest and greatest technologies;

- Develop integrated power system models on RMS and EMT platforms;
- Perform a range of power system studies from BESS optimal sizing to transients;
- Support project commissioning through SIL/HIL testing, FAT/SAT, capacity and efficiency testing, integrated systems testing, and utility witness testing. This will be part of larger team of experts and only focused on controls system.
- Support team work and project delivery. Deliver clear reports and client presentations, manage assigned scope, schedule, budget, and risks, and contribute to proposals and knowledge sharing across Jacobs. #epjobs #naepjobs

Here's what you'll need

Bachelor’s degree in Electrical Engineering or a closely related discipline, with a strong foundation in power systems, circuit theory, power electronics, and controls.

- At least five years of relevant experience in BESS, power systems engineering, grid integration, microgrids, or mission-critical electrical infrastructure, including direct involvement in BESS projects.
- Direct experience in controls hierarchy and specifications of utility scale plants, BESS sizing, performance assessment and electrical integration. Understanding of UPS operating modes, onsite generation, protective relaying, utility interfaces, and their influence on facility performance during normal operation, disturbances, and transfers.
- Hands-on power system and controls modeling and analysis experience using tools such as PSCAD/EMTDC, PSS®E, ETAP, or equivalent platforms.
- Working knowledge of applicable NEC, NESC, IEEE, IEC, NFPA, utility, and regional interconnection requirements, together with familiarity with NFPA 855, UL 9540, and UL 9540A fire-test evidence.
- Strong analytical, troubleshooting, technical writing, and presentation skills, with the ability to communicate engineering decisions to technical and nontechnical stakeholders.
- Ability to coordinate multidisciplinary teams, OEMs, consultants, and contractors across concurrent projects while maintaining safety, quality, and delivery commitments.
- Willingness to travel for client meetings, factory testing, commissioning, and project-site activities.

Ideally, you’ll also have

- A master’s degree or PhD in Electrical Engineering, Power Systems, Power Electronics, or Controls, and professional registration such as PE or PEng.
- Priority will be given to candidates from a BESS OEM, or bringing recent BESS OEM experience, particularly those with experience in power systems engineering, applications engineering, inverter or plant controls, grid integration, product validation, or commissioning.
- SIL/CHIL experience using RTDS, OPAL-RT, or Typhoon HIL; familiarity with RTAC-based integration, network protocols, oscilloscopes, and hardware/software debugging.
- Experience with SCADA, EMS, and electrical monitoring interfaces, including IEC 61850, signal lists, data models, command arbitration, and protection–control

Posted Salary Range: Minimum

145,200.00

Posted Salary Range: Upper

199,700.00

Our health and welfare benefits are designed to invest in you, and in the things you care about. Your health. Your well-being. Your security. Your future. Employees have access to medical, dental, vision, and basic life insurance, a 401(k) plan, and the ability to purchase company stock at a discount. Eligible employees may also enroll in a deferred compensation plan or the Executive Deferral Plan. Jacobs has an unlimited U.S. Personalized Paid Time Off (PPTO) policy for full-time salaried/exempt employees, seven paid holidays, and caregiver leave. And certain roles may be eligible for additional rewards, including merit increases, performance discretionary bonus, and stock.

The base salary range for this position is $145,200.00 to $199,700.00. Within the range, individual pay is determined by work location and additional factors, including job-related skills, experience, and relevant education or training.

All qualified applicants will receive consideration for employment without regard to race, color, religion, sex, sexual orientation, gender identity, national origin, disability or protected veteran status. Job posted on October 14, 2026. This position will be open for at least 3 days.

At Jacobs, we’re partnering across the globe to create the best project outcomes by maximizing the design, digital technology, and support capabilities of our Global Integrated Delivery (GID) teammates. By joining Jacobs, you’ll commit to supporting and engaging with these teams, as we work to build a company like no other.

We value collaboration and believe that in-person interactions are crucial for both our culture and client delivery. We empower employees with our hybrid working policy, allowing them to split their work week between Jacobs offices/projects and remote locations enabling them to deliver their best work.

Your application experience is important to us, and we’re keen to adapt to make every interaction even better. If you require further support or reasonable adjustments with regards to the recruitment process (for example, you require the application form in a different format), please contact the team via Careers Support .

Locations

- City State Country
- Miami Florida United States
- Irvine California United States
- Fort Worth Texas United States
- Chicago Illinois United States
- Dallas Texas United States
- Boston Massachusetts United States